

Findings — does quantum help?

QF-AgentOS — An honest benchmark and evidence harness for quantum finance

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Short answer: no. Not on any problem this platform can run, and not by a small margin. This document is the evidence, and — more usefully — the *reasons*.

Everything here is reproducible: `qf-agent arena` regenerates the leaderboard, and a regression test (`test_full_suite_has_no_quantum_advantage`) fails the build if the platform ever starts claiming otherwise.

1. The leaderboard

```
$ qf-agent arena
7 problems · 0 quantum advantage · 1 parity · 6 classical preferred.
```

problem	family	task	classical	quantum	verdict
abstention-demo	collateral_allocation	optimization	MILP 3,550	none	QUANTUM NOT FEASIBLE ON PRESENT HARDWARE
collateral-allocation	collateral_allocation	optimization	MILP 7,140	QAOA — <i>(infeasible)</i>	CLASSICAL PREFERRED
payment-routing	payment_routing	optimization	MILP 1,869	QAOA — <i>(infeasible)</i>	CLASSICAL PREFERRED
settlement-netting	settlement_netting	optimization	MILP 0	QAOA 0	QUANTUM PARITY
fraud-detection	fraud_detection	classification	AUC 0.9826	quantum kernel AUC 0.5568	CLASSICAL PREFERRED
rfq-fill	rfq_fill	classification	AUC 0.945	quantum kernel AUC 0.4964	CLASSICAL PREFERRED
infeasible	collateral_allocation	optimization	infeasible	none	CLASSICAL PREFERRED

Score convention: objective = lower better; AUC = higher better. Deterministic per seed.

Note what “—” means for quantum: the QAOA *did* run and *did* decode a solution — it was **infeasible** against the exact constraints, so it earns no score. That is the verification agent doing its job.

2. Why this is trustworthy

A negative result is only worth something if the comparison was fair. The guardrails:

- **The classical baseline is not a strawman.** Exact binary MILP via SciPy/HiGHS (plus an LP lower bound), solving these instances in ~ 1 ms. For classification, logistic regression *and* RBF kernel-ridge — with the quantum kernel judged against the **same** kernel-ridge learner, isolating the kernel.
- **Every quantum solution is mechanically re-verified** against the full exact constraint set — never trusted from the QUBO energy.
- **Statistical honesty in classification** — temporal holdout, data-leakage checks, bootstrap significance vs the RBF comparator.
- **Abstention is a first-class outcome** — the planner refuses when quantum isn't warranted.
- **The auditor is biased toward classical** — quantum must be *verified feasible* and at least match the classical optimum to earn better than “classical preferred”.
- **Determinism** — same spec + seed $\Rightarrow$ byte-identical evidence_digest.

3. Why quantum loses — per family

The interesting part. **In most cases the encoding decided the outcome before the solver ever ran.**

collateral_allocation / payment_routing — the QUBO drops or distorts constraints

The QUBO encodes the objective plus *some* constraints via penalties/slack; others (concentration, HQLA, route capacity, approval floors) are **verification-only**. The QAOA optimum therefore breaches the real constraint set, and the verifier catches it. Classical MILP handles all constraints exactly, in milliseconds.

settlement_netting — a structural impossibility result

The most technically interesting finding. Liquidity ($\text{net outflow} \leq \text{balance}$) is a **packing constraint**. Settling an obligation of amount a that overshoots a payer's headroom by δ gains reward $\sim a$ but costs only the quadratic penalty $A \cdot \delta^2$. Because δ can be arbitrarily small while a is large, **no tractable penalty weight A makes the QUBO's ground state feasible** — you would need $A \sim \text{max_amount} \cdot \text{total}$, which swamps the objective and destroys the QAOA landscape.

An adversarial sweep of **320 random instances** confirmed it: 25–90% of QUBO ground states are liquidity-infeasible, **and every single one was caught and reported honestly** (zero laundered as feasible). Parity here means QAOA matched the MILP on a benign instance — not that it is competitive.

This is not a tuning failure. It is a property of the problem class. Some financial constraints simply do not survive QUBO encoding at usable penalty weights.

fraud_detection / rfq_fill — the quantum kernel doesn't separate the data

The ZZ-feature-map fidelity kernel scores AUC **0.557** and **0.496** — the latter is *worse than a coin flip* — against classical **0.983** / **0.945** on identical data, the same learner, and the same temporal holdout. The bootstrap significance test finds no quantum improvement. There is no subtlety here: it just doesn't work.

4. Real quantum hardware (IBM)

A full 9-step pipeline run on **ibm_kingston** (IBM Quantum, Open plan):

Signal	Value
Steps / errors	9 / 0
Backend	real hardware · ibm_kingston
QPU access time	3.0 s
Reached QUBO optimum	yes (hardware sampled the ground-state energy)
Quantum contribution	yes (shot distribution beat uniform random)
Decoded solution	infeasible (violated coverage)
Verdict	CLASSICAL PREFERRED

Read that carefully, because it is the whole thesis in one run: **the quantum hardware genuinely did something — it beat random sampling and found the QUBO's optimum — and it still lost**, because the QUBO's optimum is not a feasible answer to the real problem. Good quantum execution of a lossy encoding is still a wrong answer.

Getting there required fixing three real bugs the live run exposed (a retired IBM API channel, the free plan forbidding session mode, and a verification crash on the hardware's shot histogram). All are fixed; see [real-qpu.md](#).

5. Research-grade extras

Quantum Amplitude Estimation (qf-agent estimate)

QAE has a **proven quadratic speedup** over Monte Carlo — $O(1/\epsilon)$ vs $O(1/\epsilon^2)$ queries (Brassard et al. 2002). Our MLAE implementation recovers the exact expectation to $\sim 1e-3$ and the amplitude oracle is validated to machine precision. And yet:

- At runnable sizes, **exact summation is instant and error-free**, and classical MC at the same query budget **matches or beats** the QAE estimate.
- **State preparation costs $O(2^m)$ gates** — the binding constraint. The speedup survives only if amortised across many pricings (Stamatopoulos et al. 2020).
- A useful advantage needs **early fault tolerance** (Chakrabarti et al. 2021) — a 10+ year horizon.

Verdict: CLASSICAL PREFERRED. The speedup is real, proven, and irrelevant today.

Tensor-network baseline (qf-agent simulability)

Measures whether the QAOA circuit is classically simulable by an MPS. On these dense QUBOs the output states are *highly* entangled (13-qubit collateral: entropy ≈ 5.2 bits, bond dimension ≈ 43 vs an exact-rank max of 64), so an MPS gives **no compression** — reported honestly as a small-instance artefact, **not** as evidence of hardware advantage. (Exact statevector already solves it classically.)

Methodology + citations: [research-note-quantum-extras.md](#).

6. Limitations — what these findings do not prove

Honesty cuts both ways. This platform **does not** show that quantum computing is useless in finance forever. It shows:

- **At these scales** ($\leq \sim 22$ qubits, reduced research instances), classical wins — which is unsurprising, since exact methods solve these instances outright. The platform **cannot** locate a crossover at production scale; that question stays open.
- **For these encodings.** A better formulation (not a better solver) is where any future quantum result would have to come from. The settlement finding suggests some constraint classes are structurally hostile to QUBO — a testable, generalisable claim.
- **For these techniques** (QAOA, quantum kernels, QAE). Fault-tolerant algorithms are out of reach of any simulator.

What it *does* establish, reproducibly: **the popular near-term quantum-finance demonstrations do not survive a fair comparison** — and the reason is usually the encoding, not the hardware.

7. Reproducing this

```
pip install -e ".[all,dev]"
qf-agent arena --out arena/           # the leaderboard above
qf-agent solve examples/settlement-netting.yaml
qf-agent estimate --qubits 4          # QAE vs classical MC
qf-agent simulability examples/collateral-allocation.yaml
pytest -q                             # 229 tests, incl. the no-advantage regression test
```

Every run writes an evidence bundle (manifest.json, report.md, model_card.md) with a deterministic digest. Same spec + seed $\Rightarrow$ identical digest.

Research artifact — decision-support only, not investment advice.